

Wave and scattering operators for Schrödinger operators on a one-dimensional lattice with varying spectral multiplicity

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Abstract

In this paper, we investigate Schrödinger operators on the Hilbert space of vector-valued square-integrable functions over the integers. The operators under consideration are sums of a weighted discrete Laplacian and a matrix-valued potential with sufficiently fast decay. We prove the absence of singular continuous spectrum and compute the wave and scattering operators. We show that the scattering matrix coincides with a unitary matrix introduced in a previous work. To establish these results, novel techniques are needed, since in our model, the dimension of the scattering matrix varies with the energy.

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Introduction

We study scattering theory for a class of matrix-valued Schrödinger operators on $\mathbb{Z}$ of the form

$$H = H_0 + V,$$

where the free Hamiltonian H_0 is an energy shift of the discrete Laplace operator composed with the left multiplication by an invertible matrix A and the potential V is a self-adjoint matrix multiplication operator satisfying a suitable decay condition (cf. Section 1 for a detailed definition). This model is widely used in the context of low-energy phenomena in solid-state physics.

The novelty of this model is that if the matrix A is not a scalar multiple of a unitary matrix, the dimension of the scattering matrix varies with the energy. Such models have recently been of interest, for example, in [APRR24], although there, in a similar model, only the case of a potential V supported in a single point has been studied.

A construction of the corresponding scattering matrix was given in the case $A = \mathbf{1}$ in [BFGS22] and [BFNS24], and for the general case in [BS26b]. Even though these papers establish some properties of the constructed scattering matrices, a formal proof that the matrices defined by them do, in fact, coincide with the fibration of the scattering operator with respect to the energy of the free Hamiltonian has not been provided. The present paper closes this gap.

Our proof of this result is loosely based on the ideas presented in [RS79, Chap. XI.6]. However, since, contrary to our discrete model, [RS79, Chap. XI.6] considers a model on a continuous space, we had to make some substantial adaptations. An additional difficulty, requiring new techniques, arises from the jumps in the dimension of the scattering matrix along the spectrum of the free Hamiltonian.

Structure of the paper:

In Section 1, we formally introduce the model and prove existence and completeness of wave operators $\Omega_{\pm}$ and scattering operator $S = \Omega_{-}^{*} \Omega_{+}$ (Proposition 1.2).

In Section 2, we introduce the concept of general eigenfunctions for both the free Hamiltonian H_0 and the interaction Hamiltonian H . Further, in both cases we describe bases of the space of bounded generalized eigenfunctions.

In Section 3, we use these bounded generalized eigenfunctions to define generalized Fourier transforms for the free and the interaction Hamiltonian. The construction of the generalized Fourier transform $\mathcal{F}_0$, expanding a vector in terms of the bounded generalized eigenfunctions of H_0 , is done in Section 3.1. The more elaborate construction of the generalized Fourier transforms $\mathcal{F}_{\pm}$, expanding a vector in terms of the bounded generalized eigenfunctions of H , is done in Section 3.2, by exploiting the limiting absorption principle established in [BS26b].

In Section 4, we then show that the wave operators are given by

$$\Omega_{\pm} = \mathcal{F}_{\pm}^{*} \mathcal{F}_0$$

(Theorem 4.1). Further, we prove that

$$\mathcal{F}_0 H_0 \mathcal{F}_0^{*} = \int_{\sigma(H_0)}^{\oplus} E \mathbf{1} dE \quad \text{and} \quad \mathcal{F}_0 S \mathcal{F}_0^{*} = \int_{\sigma(H_0)}^{\oplus} S^E dE$$

(Theorem 3.1.4 and Theorem 4.4(b)), thereby showing that the function $\sigma(H_0) \ni E \mapsto S^E$, constructed in [BS26b], coincides with the fiber decomposition of the scattering operator $S = \Omega_-^* \Omega_+$ with respect to the energy $E \in \sigma(H_0)$.

1 The model

Given $L \in \mathbb{N}$ and an invertible normal matrix $A \in \text{GL}_L(\mathbb{C})$, we define the free Hamiltonian H_0 on $\mathcal{H} := \ell^2(\mathbb{Z}, \mathbb{C}^L)$ by

$$(H_0\phi)(n) := A^*\phi(n+1) + A\phi(n-1), \quad n \in \mathbb{Z}.$$

Further, we consider a potential

$$(V\phi)(n) := V(n)\phi(n), \quad n \in \mathbb{Z}, \phi \in \mathcal{H},$$

where $V(n) \in \mathbb{C}^{L \times L}$ are self-adjoint $(L \times L)$ -matrices satisfying the decay condition

$$\sum_{n \in \mathbb{Z}} \|V(n)\| |n| \tau_A^{|n|} < \infty, \quad (1)$$

where

$$\tau_A := \eta_A + \sqrt{\eta_A^2 - 1} \quad \text{with} \quad \eta_A := \|A\| \cdot \|A^{-1}\|.$$

(This decay condition is optimal, see [BS26a, TODO].) We define the interaction Hamiltonian

$$H := H_0 + V.$$

By [BS26a, TODO], after conjugation by a suitable unitary operator, we can, without loss of generality, assume that

$$A = \text{diag}(a_1, \dots, a_L) \quad \text{with} \quad \{a_1, \dots, a_L\} = \sigma(A)$$

and

$$a_1 \geq \dots \geq a_L > 0. \quad (2)$$

From now on, for the rest of the paper, we will assume that A is of this form. Given this normal form for the matrix A , we can prove the following proposition:

Proposition 1.1. *The operator H_0 has purely absolutely continuous spectrum and*

$$\sigma(H_0) = [-2\|A\|, 2\|A\|].$$

Proof. We consider the operator h_0 on $\ell^2(\mathbb{Z}, \mathbb{C})$ defined by

$$(h_0\phi)(n) := \phi(n+1) + \phi(n-1).$$

Then

$$H_0 = \bigoplus_{l=1}^L a_l h_0.$$

By [BFGS22, Sec. 1.1.1], we have

$$\sigma(h_0) = \sigma_{\text{ac}}(h_0) = [-2, 2],$$

so

$$\begin{aligned}\sigma(H_0) &= \sigma_{\text{ac}}(H_0) = \sigma\left(\bigoplus_{l=1}^L a_l h_0\right) = \bigcup_{l=1}^L a_l \sigma(h_0) = \bigcup_{l=1}^L a_l [-2, 2] \\ &= \bigcup_{l=1}^L [-2a_l, 2a_l] = [-2a_1, 2a_1] = [-2\|A\|, 2\|A\|],\end{aligned}$$

using (2) in the penultimate step. $\square$

We next observe the existence of the wave operators in our model:

Proposition 1.2. *The wave operators*

$$\Omega_{\pm} := \text{s-}\lim_{t \rightarrow \mp\infty} e^{itH} e^{-itH_0}$$

exist and are complete, i.e. partial isometries from $P_{\text{ac}}(H_0)\mathcal{H} = \mathcal{H}$ to $P_{\text{ac}}(H)\mathcal{H}$.

Proof. The operator $H - H_0 = V$ is of trace class, since, by assumption (1), we have

$$\|V\|_1 = \sum_{n \in \mathbb{Z}} \|V(n)\|_1 \leq \sum_{n \in \mathbb{Z}} L \|V(n)\| \leq L \cdot \left[\|V(0)\| + \sum_{n \in \mathbb{Z}} \|V(n)\| |n| \tau_A^{|n|} \right] < \infty.$$

The existence and completeness of the wave operators then follow by the Kato–Rosenblum Theorem ([RS79, Thm. XI.8]). $\square$

The remainder of the paper is devoted to computing these wave operators and the scattering operator $S = \Omega_-^* \Omega_+ \in U(\mathcal{H})$.

2 Generalized eigenfunctions

In this section, we introduce generalized eigenfunctions and present facts about them that we will use in later sections to calculate the wave and scattering operators. We begin with the definition:

Definition 2.1. We write $(\mathbb{C}^L)^{\mathbb{Z}}$ for the set of all functions $\phi : \mathbb{Z} \rightarrow \mathbb{C}^L$ and consider the operators

$$H_0 : (\mathbb{C}^L)^{\mathbb{Z}} \rightarrow (\mathbb{C}^L)^{\mathbb{Z}}, \quad (H_0 \phi)(n) = A^* \phi(n+1) + A \phi(n-1)$$

and

$$H : (\mathbb{C}^L)^{\mathbb{Z}} \rightarrow (\mathbb{C}^L)^{\mathbb{Z}}, \quad (H \phi)(n) = A^* \phi(n+1) + V(n) \phi(n) + A \phi(n-1).$$

Then, for $K \in \{H_0, H\}$ and $E \in \mathbb{C}$, we define the space of *generalized eigenfunctions*

$$\mathcal{G}(K, E) := \{\phi \in (\mathbb{C}^L)^{\mathbb{Z}} : K\phi = E\phi\}.$$

Further, we set

$$\ell^\infty(K, E) := \mathcal{G}(K, E) \cap \ell^\infty(\mathbb{Z}, \mathbb{C}^L) = \left\{ \phi \in \mathcal{G}(K, E) : \sup_{n \in \mathbb{Z}} \|\phi(n)\| < \infty \right\}.$$

In the case $K = H_0$, the space $\ell^\infty(H_0, E)$ can be computed explicitly. To this end, we introduce the following definition:

Definition 2.2. For $E \in \mathbb{R}$, we define

$$B^E := \{l \in \{1, \dots, L\} : E \in [-2a_l, 2a_l]\}.$$

Further, by $\arccos$ we denote the inverse of the function

$$\cos : [0, \pi] \rightarrow [-1, 1].$$

Then, for $l \in B^E$ and $\sigma \in \{+, -\}$, we define $v_l^{E, \sigma} \in (\mathbb{C}^L)^\mathbb{Z}$ by

$$v_l^{E, \sigma}(n) := \sigma \left(e^{-i\sigma \arccos\left(\frac{E}{2a_l}\right)} \right)^n \cdot \sqrt{-\frac{d}{dE} \arccos\left(\frac{E}{2a_l}\right)} \cdot e_l,$$

denoting by e_l the l -th standard basis vector, i.e. the vector having a 1 in the l -th entry and having all other entries equal to 0.

Proposition 2.3. ([BS26b, TODO]) *Let $E \in \mathbb{R} \setminus \{\pm 2a_1, \dots, \pm 2a_L\}$. Then*

$$\left\{ v_l^{E, \sigma} : l \in B^E, \sigma \in \{+, -\} \right\}$$

is a basis of the $2|B^E|$ -dimensional space $\ell^\infty(H_0, E)$.

The space $\ell^\infty(H, E)$ is characterized by the following theorem:

Theorem 2.4. ([BS26b, TODO]) *There exists a set $\mathfrak{D} \subseteq \mathbb{R}$ such that the following hold:*

- (a) $\{\pm 2a_1, \dots, \pm 2a_L\} \cup \sigma_{\text{pp}}(H) \subseteq \mathfrak{D}$.
- (b) $\overline{\mathfrak{D}}$ is at most countable.
- (c) For every $E \in \mathbb{R} \setminus \mathfrak{D}$, $l \in B^E$, $\sigma \in \{+, -\}$, and $n \in \mathbb{Z}$, the limits

$$\begin{aligned} w_{l, \pm}^{E, \sigma}(n) &:= \lim_{\varepsilon \downarrow 0} \left((H - E \mp i\varepsilon)^{-1} (H_0 - E \mp i\varepsilon) v_l^{E, \sigma} \right)(n) \\ &= v_l^{E, \sigma}(n) - \lim_{\varepsilon \downarrow 0} \left((H - E \mp i\varepsilon)^{-1} V v_l^{E, \sigma} \right)(n) \end{aligned}$$

exist and define functions $w_{l, \pm}^{E, \sigma} \in \ell^\infty(H, E)$.

- (d) For every $E \in \mathbb{R} \setminus \mathfrak{D}$, the space $\ell^\infty(H, E)$ is $2|B^E|$ -dimensional and each of the following sets is a basis of it:

$$\left\{ w_{l, +}^{E, \sigma} : l \in B^E, \sigma \in \{+, -\} \right\} \quad \text{and} \quad \left\{ w_{l, -}^{E, \sigma} : l \in B^E, \sigma \in \{+, -\} \right\}.$$

- (e) For every $E \in \mathbb{R} \setminus \mathfrak{D}$, $l \in B^E$, $\sigma \in \{+, -\}$, and $n \in \mathbb{Z}$, one has

$$\lim_{\varepsilon \downarrow 0} \left((H_0 - E \mp i\varepsilon)^{-1} V w_{l, \pm}^{E, \sigma} \right)(n) = v_l^{E, \sigma}(n) - w_{l, \pm}^{E, \sigma}(n).$$

Remark 2.5. We shall repeatedly use the identity

$$\lim_{\varepsilon \downarrow 0} \left((H - E \mp i\varepsilon)^{-1} V v_l^{E, \sigma} \right)(n) = v_l^{E, \sigma}(n) - w_{l, \pm}^{E, \sigma}(n), \quad (3)$$

which is an immediate consequence of Theorem 2.4(c).

3 Generalized Fourier transforms

Throughout this section, we consider the unitary Fourier transform

$$\mathcal{F} : \ell^2(\mathbb{Z}, \mathbb{C}^L) \rightarrow L^2([-\pi, \pi], \mathbb{C}^L)$$

defined by

$$(\mathcal{F}f)(\theta) := \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} e^{in\theta} f(n), \quad f \in \ell^1(\mathbb{Z}, \mathbb{C}^L).$$

We will construct variants of this Fourier transform, replacing the basis $(e^{in\theta})_{n \in \mathbb{Z}}$ by the functions $(v_l^{E,\sigma}(n))_{n \in \mathbb{Z}}$ in case of the free Hamiltonian and respectively $(w_{l,\pm}^{E,\sigma}(n))_{n \in \mathbb{Z}}$ in case of the interaction Hamiltonian. We will later use these generalized Fourier transforms to calculate the wave and scattering operators. We start with the free Hamiltonian:

3.1 A Fourier transform for the free Hamiltonian

For the free Hamiltonian, instead of the Hilbert space $L^2([-\pi, \pi], \mathbb{C}^L)$, the image of our generalized Fourier transform will be the following Hilbert space:

Definition 3.1.1. We consider the Hilbert space

$$\mathcal{K} := \bigoplus_{\substack{l \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} L^2([-2a_l, 2a_l], \mathbb{C})$$

endowed with the scalar product

$$\langle f|g \rangle := \int^{\mathcal{K}} \overline{f(E)} g(E) dE := \sum_{\substack{l \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \int_{-2a_l}^{2a_l} \overline{f_{l,\sigma}(E)} g_{l,\sigma}(E) dE.$$

Further, we define the operator

$$\Phi : L^2([-\pi, \pi], \mathbb{C}^L) \rightarrow \mathcal{K}$$

by

$$(\Phi f)_{l,\sigma}(E) := \sigma f_l \left(\sigma \arccos \left(\frac{E}{2a_l} \right) \right) \cdot \sqrt{-\frac{d}{dE} \arccos \left(\frac{E}{2a_l} \right)}.$$

Lemma 3.1.2. *The map $\Phi : L^2([-\pi, \pi], \mathbb{C}^L) \rightarrow \mathcal{K}$ is unitary.*

Proof. For $f, g \in L^2([-\pi, \pi], \mathbb{C}^L)$, using the substitution $\theta = \sigma \arccos \left(\frac{E}{2a_l} \right)$, one has

$$\begin{aligned} \langle \Phi f | \Phi g \rangle &= \int^{\mathcal{K}} \overline{(\Phi f)(E)} (\Phi g)(E) dE \\ &= \sum_{\substack{l \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \int_{-2a_l}^{2a_l} \overline{f_l \left(\sigma \arccos \left(\frac{E}{2a_l} \right) \right)} g_l \left(\sigma \arccos \left(\frac{E}{2a_l} \right) \right) \cdot \left(-\frac{d}{dE} \arccos \left(\frac{E}{2a_l} \right) \right) dE \\ &= \sum_{\substack{l \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \int_{\sigma\pi}^0 \overline{f_l(\theta)} g_l(\theta) \cdot (-\sigma) d\theta = \sum_{l \in \{1, \dots, L\}} \int_{-\pi}^{\pi} \overline{f_l(\theta)} g_l(\theta) d\theta = \langle f | g \rangle, \end{aligned}$$

which shows that Φ is an isometry. Further, Φ is invertible with its inverse being given by

$$(\Phi^{-1}f)_l(\theta) = \operatorname{sgn}(\theta)f_{l,\operatorname{sgn}(\theta)}(2a_l \cos \theta) \cdot \sqrt{2a_l \sin |\theta|}$$

which implies that Φ is unitary. $\square$

We now define our generalized Fourier transform, of which we will see that it can be calculated using the generalized eigenfunctions of the free Hamiltonian:

Definition 3.1.3. We define

$$\mathcal{F}_0 := \Phi \circ \mathcal{F} : \ell^2(\mathbb{Z}, \mathbb{C}^L) \rightarrow \mathcal{K}.$$

Theorem 3.1.4. *The operator $\mathcal{F}_0$ satisfies the following properties:*

(a) $\mathcal{F}_0$ is unitary.

(b) For every $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$, $l = 1, \dots, L$, and $\sigma \in \{+, -\}$, one has

$$(\mathcal{F}_0 f)_{l,\sigma}(E) = \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle v_l^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L}, \quad E \in [-2a_l, 2a_l].$$

(c) $\mathcal{F}_0 H_0 = M_{\operatorname{Id}} \mathcal{F}_0$.

Proof. (a) The unitarity follows by

$$\mathcal{F}_0 = \Phi \circ \mathcal{F}$$

and the fact that $\mathcal{F}$ and Φ are unitary by Lemma 3.1.2.

(b) For $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$, by Definition 3.1.1 and Definition 2.2, one has

$$\begin{aligned} (\mathcal{F}_0 f)_{l,\sigma}(E) &= \sigma(\mathcal{F}f)_l \left(\sigma \arccos \left(\frac{E}{2a_l} \right) \right) \cdot \sqrt{-\frac{d}{dE} \arccos \left(\frac{E}{2a_l} \right)} \\ &= \sigma \left(\frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} e^{in\sigma \arccos \left(\frac{E}{2a_l} \right)} f_l(n) \right) \cdot \sqrt{-\frac{d}{dE} \arccos \left(\frac{E}{2a_l} \right)} \\ &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle \sigma \left(e^{-i\sigma \arccos \left(\frac{E}{2a_l} \right)} \right)^n \cdot \sqrt{-\frac{d}{dE} \arccos \left(\frac{E}{2a_l} \right)} \cdot e_l \middle| f(n) \right\rangle_{\mathbb{C}^L} \\ &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle v_l^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L}. \end{aligned}$$

(c) By (b), for $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$, $l = 1, \dots, L$ and $\sigma \in \{+, -\}$, we have

$$\begin{aligned} (\mathcal{F}_0 H_0 f)_{l,\sigma}(E) &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle v_l^{E,\sigma}(n) \middle| (H_0 f)(n) \right\rangle_{\mathbb{C}^L} \\ &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle \left(H_0 v_l^{E,\sigma} \right)(n) \middle| f(n) \right\rangle_{\mathbb{C}^L} \\ &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle E \cdot v_l^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L} \\ &= E \cdot (\mathcal{F}_0 f)_{l,\sigma}(E) = (M_{\operatorname{Id}} \mathcal{F}_0 f)_{l,\sigma}(E). \end{aligned}$$

The statement then follows, since $\ell^1(\mathbb{Z}, \mathbb{C}^L) \subseteq \ell^2(\mathbb{Z}, \mathbb{C}^L)$ is dense. $\square$

We finish this section by a technical lemma that will be needed later for the construction of generalized Fourier transforms using eigenfunctions of the interaction Hamiltonian. For this lemma, we will need the *Green's function* $G_K^{E,\sigma}$ as defined in the following theorem:

Theorem 3.1.5. ([BS26b, TODO, TODO]) *Let $K \in \{H_0, H\}$ and $\sigma \in \{+, -\}$. Then, for every $E \in \overline{\mathbb{C}_\sigma} \setminus \mathfrak{D}$, there exists a function*

$$G_K^{E,\sigma} : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{C}^{L \times L}$$

such that the following hold:

- (a) *For every $E \in \overline{\mathbb{C}_\sigma} \setminus \mathfrak{D}$ and $j \in \mathbb{Z}$, one has $G_K^{E,\sigma}(\cdot, j) \in \ell^\infty(\mathbb{Z}, \mathbb{C}^{L \times L})$.*
- (b) *For every compact subset $F \subseteq \overline{\mathbb{C}_\sigma} \setminus \mathfrak{D}$ and $j \in \mathbb{Z}$, one has*

$$\sup_{E \in F} \left\| G_K^{E,\sigma}(\cdot, j) \right\|_\infty < \infty.$$

- (c) *The function*

$$\overline{\mathbb{C}_\sigma} \setminus \mathfrak{D} \ni E \mapsto V(\cdot) G_K^{E,\sigma}(\cdot, j) \in \ell^1(\mathbb{Z}, \mathbb{C}^{L \times L})$$

is continuous.

- (d) *If $E \notin \sigma(K)$, for every $f \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$, one has*

$$(K - E)^{-1} f = \sum_{j \in \mathbb{Z}} G_K^{E,\sigma}(\cdot, j) f(j).$$

Lemma 3.1.6. *Let $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$. For $E \in \overline{\mathbb{C}_\pm} \setminus \mathfrak{D}$, we set*

$$f_\pm^{E,g} := V \sum_{j \in \mathbb{Z}} G_H^{E,\pm}(\cdot, j) g(j) \in \ell^1(\mathbb{Z}, \mathbb{C}^L) \quad \text{and} \quad F_\pm^{E,g} := \mathcal{F}_0 \left(g - f_\pm^{E,g} \right) \in \mathcal{K}.$$

Further, let $l \in \{1, \dots, L\}$ and $\sigma \in \{+, -\}$ and let $I \subseteq \mathbb{R} \setminus \mathfrak{D}$ and $J \subseteq (-2a_l, 2a_l)$ be compact subsets. Then, the following hold:

- (a) *The map*

$$\overline{\mathbb{C}_\pm} \setminus \mathfrak{D} \ni E \mapsto \left(F_\pm^{E,g} \right)_{l,\sigma} \Big|_J \in C(J, \mathbb{C})$$

is continuous.

- (b) *One has*

$$\sup_{\substack{E \in I \\ X \in J}} \left| \left(F_\pm^{E \pm i\varepsilon, g} - F_\pm^{E, g} \right)_{l,\sigma}(X) \right| \xrightarrow{\varepsilon \downarrow 0} 0.$$

- (c) *One has*

$$\lim_{\delta \downarrow 0} \sup_{\substack{E \in I \cap [-2a_l, 2a_l] \\ X \in [E - \delta, E + \delta]}} \left| \left(F_\pm^{E, g} \right)_{l,\sigma}(X) - \left(F_\pm^{E, g} \right)_{l,\sigma}(E) \right| = 0.$$

- (d) *For every compact subset $F \subseteq \overline{\mathbb{C}_\pm} \setminus \mathfrak{D}$, one has*

$$\sup_{E \in F} \left\| F_\pm^{E, g} \right\|_{\mathcal{K}} < \infty.$$

Proof. (a) By Theorem 3.1.5(c), the function

$$\overline{\mathbb{C}}_{\pm} \setminus \mathfrak{D} \ni E \mapsto f_{\pm}^{E,g} \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$$

is continuous. Since

$$\left(F_{\pm}^{E,g} \right)_{l,\sigma} \Big|_J = \left(\mathcal{F}_0 \left(g - f_{\pm}^{E,g} \right) \right)_{l,\sigma} \Big|_J = \left(\Phi \mathcal{F} \left(g - f_{\pm}^{E,g} \right) \right)_{l,\sigma} \Big|_J$$

and the Fourier transform

$$\mathcal{F} : \ell^1(\mathbb{Z}, \mathbb{C}^L) \rightarrow C([-\pi, \pi], \mathbb{C}^L)$$

is continuous, it suffices to show that the function

$$C([-\pi, \pi], \mathbb{C}^L) \ni f \mapsto (\Phi f)_{l,\sigma} \Big|_J \in C(J, \mathbb{C})$$

is continuous. This follows, since, for $f \in C([-\pi, \pi], \mathbb{C}^L)$, we have

$$\begin{aligned} \left\| (\Phi f)_{l,\sigma} \Big|_J \right\|_{\infty} &= \sup_{X \in J} \left| \sigma f_l \left(\sigma \arccos \left(\frac{X}{2a_l} \right) \right) \cdot \sqrt{-\frac{d}{dE} \arccos \left(\frac{E}{2a_l} \right)} \right| \\ &\leq \sup_{X \in J} \left| \sqrt{-\frac{d}{dE} \arccos \left(\frac{E}{2a_l} \right)} \right| \cdot \|f\|_{\infty} \end{aligned}$$

and the fact that the supremum is finite due to the compactness of J .

(b) We consider the compact set

$$F := I + \sigma i[0, 1].$$

By (a), the function

$$F \ni E \mapsto \left(F_{\pm}^{E,g} \right)_{l,\sigma} \Big|_J \in C(J, \mathbb{C})$$

is continuous and therefore, by the Heine–Cantor Theorem, uniformly continuous. This immediately yields

$$\begin{aligned} &\sup_{\substack{E \in I \\ X \in J}} \left| \left(F_{\pm}^{E \pm i\varepsilon, g} - F_{\pm}^{E,g} \right)_{l,\sigma} (X) \right| \\ &= \sup_{E \in I} \left\| \left(F_{\pm}^{E \pm i\varepsilon, g} \right)_{l,\sigma} \Big|_J - \left(F_{\pm}^{E,g} \right)_{l,\sigma} \Big|_J \right\|_{\infty} \xrightarrow{\varepsilon \downarrow 0} 0. \end{aligned}$$

(c) Since $I \subseteq \mathbb{R} \setminus \mathfrak{D} \subseteq \mathbb{R} \setminus \{\pm 2a_l\}$, the set $I \cap [-2a_l, 2a_l]$ is a compact subset of $(-2a_l, 2a_l)$ and therefore we can choose $r > 0$ such that

$$I_r := (I \cap [-2a_l, 2a_l]) + [-r, r] \subseteq (-2a_l, 2a_l).$$

We first notice that the function

$$I \times I_r \ni (E, X) \mapsto \left(F_{\pm}^{E,g} \right)_{l,\sigma} (X) \in \mathbb{C}$$

is continuous since, by (a), we have

$$\begin{aligned}
& \left| \left(F_{\pm}^{E,g} \right)_{l,\sigma} (X) - \left(F_{\pm}^{E',g} \right)_{l,\sigma} (X') \right| \\
& \leq \left| \left(F_{\pm}^{E,g} \right)_{l,\sigma} (X) - \left(F_{\pm}^{E,g} \right)_{l,\sigma} (X') \right| + \left| \left(F_{\pm}^{E,g} \right)_{l,\sigma} (X') - \left(F_{\pm}^{E',g} \right)_{l,\sigma} (X') \right| \\
& \leq \left| \left(F_{\pm}^{E,g} \right)_{l,\sigma} (X) - \left(F_{\pm}^{E,g} \right)_{l,\sigma} (X') \right| + \left\| \left(F_{\pm}^{E,g} - F_{\pm}^{E',g} \right)_{l,\sigma} \right\|_{I_r} \\
& \xrightarrow{(E',X') \rightarrow (E,X)} 0.
\end{aligned}$$

As a continuous function on a compact set, by the Heine–Cantor Theorem, it is uniformly continuous. This implies

$$\lim_{\delta \downarrow 0} \sup_{\substack{(E,X), (E',X') \in I \times I_r \\ (E-E')^2 + (X-X')^2 \leq \delta^2}} \left| \left(F_{\pm}^{E,g} \right)_{l,\sigma} (X) - \left(F_{\pm}^{E',g} \right)_{l,\sigma} (X') \right| = 0.$$

The statement then follows, since

$$\begin{aligned}
& \{((E, X), (E, E)) : E \in I \cap [-2a_l, 2a_l], X \in [E - \delta, E + \delta]\} \\
& \subseteq \{((E, X), (E', X')) \in (I \times J)^2 : (E - E')^2 + (X - X')^2 \leq \delta^2\}.
\end{aligned}$$

(d) By Theorem 3.1.5(c), the function

$$\overline{\mathbb{C}_{\pm}} \setminus \mathfrak{D} \ni E \mapsto f_{\pm}^{E,g} \in \ell^1(\mathbb{Z}, \mathbb{C}^L) \subseteq \ell^2(\mathbb{Z}, \mathbb{C}^L)$$

is continuous. Since $F_{\pm}^{E,g} = \mathcal{F}_0 \left(g - f_{\pm}^{E,g} \right)$, the statement immediately follows by the continuity of $\mathcal{F}_0$ due to Theorem 3.1.4(a). $\square$

3.2 A Fourier transform for the interaction Hamiltonian

In this section, we want to define generalized Fourier transforms $\mathcal{F}_{\pm}$ analogous to $\mathcal{F}_0$ in a way such that Theorem 3.1.4(b) holds for the eigenfunctions of the interaction Hamiltonian instead of the free Hamiltonian, i.e.

$$(\mathcal{F}_{\pm} f)_{l,\sigma}(E) := \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle w_{l,\pm}^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L}, \quad f \in \ell^1(\mathbb{Z}, \mathbb{C}^L).$$

The difficulty here is to see that, defined in this way, $\mathcal{F}_{\pm} f$ is an element of $\mathcal{K}$ and that $\mathcal{F}_{\pm}$ can be extended to a unitary operator on $\ell^2(\mathbb{Z}, \mathbb{C}^L)$. We will address these issues later in the section. We start with the following definition:

Definition 3.2.1. Let $I \subseteq \mathbb{R}$. We write $\mathcal{M}(I, \mathbb{C})$ for the set of equivalence classes of Borel measurable functions up to values on a Lebesgue zero-set. Further, we define

$$\tilde{\mathcal{F}}_{\pm} : \ell^1(\mathbb{Z}, \mathbb{C}^L) \rightarrow \bigoplus_{\substack{l \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \mathcal{M}([-2a_l, 2a_l], \mathbb{C})$$

by

$$(\tilde{\mathcal{F}}_{\pm} f)_{l,\sigma}(E) := \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle w_{l,\pm}^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L}, \quad f \in \ell^1(\mathbb{Z}, \mathbb{C}^L), E \in [-2a_l, 2a_l] \setminus \mathfrak{D}.$$

(Notice that $\mathfrak{D}$ is a Lebesgue zero-set by Theorem 2.4(b).)

We now relate the so defined operators $\tilde{\mathcal{F}}_{\pm}$ to Lemma 3.1.6:

Lemma 3.2.2. *Let $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, $l \in \{1, \dots, L\}$, and $\sigma \in \{+, -\}$. Then, for every $E \in [-2a_l, 2a_l] \setminus \mathfrak{D}$, one has*

$$\left(F_{\mp}^{E,g} \right)_{l,\sigma}(E) = (\tilde{\mathcal{F}}_{\pm} g)_{l,\sigma}(E).$$

Proof. By Theorem 3.1.5(c) and Theorem 3.1.4(a), we get

$$\left(\mathcal{F}_0 f_{\mp}^{E,g} \right)_{l,\sigma}(E) = \lim_{\varepsilon \downarrow 0} \left(\mathcal{F}_0 f_{\mp}^{E \mp i\varepsilon, g} \right)_{l,\sigma}(E).$$

Using that, by Theorem 3.1.5(d), we have

$$f_{\mp}^{E \mp i\varepsilon, g} = V(H - E \pm i\varepsilon)^{-1} g,$$

this yields

$$\begin{aligned} \left(\mathcal{F}_0 f_{\mp}^{E,g} \right)_{l,\sigma}(E) &= \lim_{\varepsilon \downarrow 0} \left(\mathcal{F}_0 V(H - E \pm i\varepsilon)^{-1} g \right)_{l,\sigma}(E) \\ &= \lim_{\varepsilon \downarrow 0} \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle v_l^{E,\sigma}(n) \middle| (V(H - E \pm i\varepsilon)^{-1} g)(n) \right\rangle_{\mathbb{C}^L} \\ &= \lim_{\varepsilon \downarrow 0} \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle \left((H - E \mp i\varepsilon)^{-1} V v_l^{E,\sigma} \right)(n) \middle| g(n) \right\rangle_{\mathbb{C}^L} \\ &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle v_l^{E,\sigma}(n) - w_{l,\pm}^{E,\sigma}(n) \middle| g(n) \right\rangle_{\mathbb{C}^L} \\ &= ((\mathcal{F}_0 - \tilde{\mathcal{F}}_{\pm})g)_{l,\sigma}(E), \end{aligned}$$

using (3) in the penultimate and Theorem 3.1.4(b) in the last step. We then get

$$\begin{aligned} (\tilde{\mathcal{F}}_{\pm} g)_{l,\sigma}(E) &= (\mathcal{F}_0 g - (\mathcal{F}_0 - \tilde{\mathcal{F}}_{\pm})g)_{l,\sigma}(E) \\ &= \left(\mathcal{F}_0 g - \mathcal{F}_0 f_{\mp}^{E,g} \right)_{l,\sigma}(E) = \left(F_{\mp}^{E,g} \right)_{l,\sigma}(E). \quad \square \end{aligned}$$

Before we can use the above result to extend $\tilde{\mathcal{F}}_{\pm}$ to an operator on $\ell^2(\mathbb{Z}, \mathbb{C}^L)$, we need the following technical lemma:

Lemma 3.2.3. *Let $f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$ and let $I \subseteq \mathbb{R} \setminus \mathfrak{D}$ by a compact subset. Then, the following hold:*

(a) *For $l \in \{1, \dots, L\}$ and $\sigma \in \{+, -\}$, one has*

$$\begin{aligned} \frac{1}{\pi} \int_I \int_{[-2a_l, 2a_l]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} \overline{\left(F_{\mp}^{E \mp i\varepsilon, f} \right)_{l,\sigma}(X)} \left(F_{\mp}^{E \mp i\varepsilon, g} \right)_{l,\sigma}(X) dX dE \\ \xrightarrow{\varepsilon \downarrow 0} \int_{[-2a_l, 2a_l]} \chi_I(E) \overline{\left(\tilde{\mathcal{F}}_{\pm} f \right)_{l,\sigma}(E)} \left(\tilde{\mathcal{F}}_{\pm} g \right)_{l,\sigma}(E) dE. \end{aligned}$$

(b) *One has*

$$\begin{aligned} & \frac{\varepsilon}{\pi} \int_I \langle (H - E \pm i\varepsilon)^{-1} f | (H - E \pm i\varepsilon)^{-1} g \rangle dE \\ & \xrightarrow{\varepsilon \downarrow 0} \int^{\mathcal{K}} \chi_I(E) \overline{(\tilde{\mathcal{F}}_{\pm} f)(E)} (\tilde{\mathcal{F}}_{\pm} g)(E) dE. \end{aligned}$$

Proof. (a) Set

$$h(Y, X) := \overline{(F_{\mp}^{Y,f})_{l,\sigma}}(X) (F_{\mp}^{Y,g})_{l,\sigma}(X).$$

By Lemma 3.2.2, we have

$$h(E, E) = \overline{(F_{\mp}^{E,f})_{l,\sigma}}(E) (F_{\mp}^{E,g})_{l,\sigma}(E) = \overline{(\tilde{\mathcal{F}}_{\pm} f)_{l,\sigma}}(E) (\tilde{\mathcal{F}}_{\pm} g)_{l,\sigma}(E),$$

so, setting

$$\mathcal{I} := \int_{I \cap [-2a_l, 2a_l]} h(E, E) dE = \int_{[-2a_l, 2a_l]} \chi_I(E) h(E, E) dE$$

we have to show that

$$\frac{1}{\pi} \int_I \int_{[-2a_l, 2a_l]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE \xrightarrow{\varepsilon \downarrow 0} \mathcal{I}. \quad (4)$$

Since $I \subseteq \mathbb{R} \setminus \mathfrak{D} \subseteq \mathbb{R} \setminus \{\pm 2a_l\}$, the set $I \cap [-2a_l, 2a_l]$ is a compact subset of $(-2a_l, 2a_l)$ and therefore, for $\delta > 0$ sufficiently small, one has

$$(I \cap [-2a_l, 2a_l]) + [-\delta, \delta] \subseteq (-2a_l, 2a_l).$$

We then set

$$M_{\delta} := \{(E, X) \in I \times [-2a_l, 2a_l] : E \in I \cap [-2a_l, 2a_l], X \in [E - \delta, E + \delta]\}$$

and

$$N_{\delta} := (I \times [-2a_l, 2a_l]) \setminus M_{\delta}.$$

We write

$$\frac{1}{\pi} \int_I \int_{[-2a_l, 2a_l]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE = A_{\delta}(\varepsilon) + B_{\delta}(\varepsilon) + C_{\delta}(\varepsilon) + D_{\delta}(\varepsilon),$$

with

$$\begin{aligned} A_{\delta}(\varepsilon) &:= \frac{1}{\pi} \int_{N_{\delta}} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE, \\ B_{\delta}(\varepsilon) &:= \frac{1}{\pi} \int_{M_{\delta}} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} [h(E \mp i\varepsilon, X) - h(E, X)] dX dE, \\ C_{\delta}(\varepsilon) &:= \frac{1}{\pi} \int_{M_{\delta}} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} [h(E, X) - h(E, E)] dX dE, \\ D_{\delta}(\varepsilon) &:= \frac{1}{\pi} \int_{M_{\delta}} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E, E) dX dE. \end{aligned}$$

We will now prove (4), by showing the following:

- (1) For every $\delta > 0$, one has $\lim_{\varepsilon \downarrow 0} A_\delta(\varepsilon) = 0$.
- (2) For every $\delta > 0$, one has $\lim_{\varepsilon \downarrow 0} B_\delta(\varepsilon) = 0$.
- (3) One has $\limsup_{\delta \downarrow 0} \limsup_{\varepsilon > 0} |C_\delta(\varepsilon)| = 0$.
- (4) For every $\delta > 0$, one has $\lim_{\varepsilon \downarrow 0} D_\delta(\varepsilon) = \mathcal{I}$.

The statement then follows since, for every $\delta > 0$, one has

$$\begin{aligned}
& \limsup_{\varepsilon \downarrow 0} \left| \frac{1}{\pi} \int_I \int_{[-2a_l, 2a_l]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE - \mathcal{I} \right| \\
&= \limsup_{\varepsilon \downarrow 0} |A_\delta(\varepsilon) + B_\delta(\varepsilon) + C_\delta(\varepsilon) + D_\delta(\varepsilon) - \mathcal{I}| \\
&\leq \limsup_{\varepsilon \downarrow 0} |A_\delta(\varepsilon)| + \limsup_{\varepsilon \downarrow 0} |B_\delta(\varepsilon)| + \limsup_{\varepsilon \downarrow 0} |C_\delta(\varepsilon)| + \limsup_{\varepsilon \downarrow 0} |D_\delta(\varepsilon) - \mathcal{I}| \\
&= \limsup_{\varepsilon \downarrow 0} |C_\delta(\varepsilon)| \leq \sup_{\varepsilon > 0} |C_\delta(\varepsilon)|
\end{aligned}$$

and therefore

$$\begin{aligned}
& \limsup_{\varepsilon \downarrow 0} \left| \frac{1}{\pi} \int_I \int_{[-2a_l, 2a_l]} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE - \mathcal{I} \right| \\
&\leq \inf_{\delta > 0} \sup_{\varepsilon > 0} |C_\delta(\varepsilon)| \leq \limsup_{\delta \downarrow 0} \sup_{\varepsilon > 0} |C_\delta(\varepsilon)| = 0.
\end{aligned}$$

We now prove the above statements:

- (1) Since $I \subseteq \mathbb{R} \setminus \mathfrak{D} \subseteq \mathbb{R} \setminus \{\pm 2a_l\}$, the set $I \setminus [-2a_l, 2a_l]$ is a compact subset of $\mathbb{R} \setminus [-2a_l, 2a_l]$ and therefore

$$r := \text{dist}(I \setminus [-2a_l, 2a_l], [-2a_l, 2a_l]) > 0.$$

For $(E, X) \in N_\delta$, we either have $E \in I \setminus [-2a_l, 2a_l]$ or $E \in I \cap [-2a_l, 2a_l]$. In the first case, we get

$$\frac{\varepsilon}{(X - E)^2 + \varepsilon^2} \leq \frac{\varepsilon}{r^2}$$

and in the second case, we have $X \notin [E - \delta, E + \delta]$, so

$$\frac{\varepsilon}{(X - E)^2 + \varepsilon^2} \leq \frac{\varepsilon}{\delta^2}.$$

Therefore, setting $C := \pi^{-1} \max\{r^{-2}, \delta^{-2}\}$, we get

$$\frac{1}{\pi} \frac{\varepsilon}{(X - E)^2 + \varepsilon^2} \leq C\varepsilon \quad \forall (E, X) \in N_\delta.$$

For $\varepsilon \leq 1$, this implies

$$\begin{aligned}
|A_\delta(\varepsilon)| &= \left| \frac{1}{\pi} \int_{N_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} h(E \mp i\varepsilon, X) dX dE \right| \\
&\leq C\varepsilon \int_{N_\delta} |h(E \mp i\varepsilon, X)| dX dE \\
&\leq C\varepsilon \int_I \int_{[-2a_l, 2a_l]} |h(E \mp i\varepsilon, X)| dX dE \\
&= C\varepsilon \int_I \int_{[-2a_l, 2a_l]} \left| \overline{\left(F_{\mp}^{E \mp i\varepsilon, f} \right)_{l, \sigma}(X)} \left(F_{\mp}^{E \mp i\varepsilon, g} \right)_{l, \sigma}(X) \right| dX dE \\
&\leq C\varepsilon \int_I \left\| F_{\mp}^{E \mp i\varepsilon, f} \right\|_{\mathcal{K}} \cdot \left\| F_{\mp}^{E \mp i\varepsilon, g} \right\|_{\mathcal{K}} dE \\
&\leq C\varepsilon \cdot \lambda_1(I) \cdot \sup_{Y \in I \mp i[0, 1]} \left\| F_{\mp}^{Y, f} \right\|_{\mathcal{K}} \cdot \left\| F_{\mp}^{Y, g} \right\|_{\mathcal{K}} \xrightarrow{\varepsilon \downarrow 0} 0,
\end{aligned}$$

denoting by λ_1 the Lebesgue measure on $\mathbb{R}$ and using that, by Lemma 3.1.6(d), we have

$$\sup_{Y \in I \mp i[0, 1]} \left\| F_{\mp}^{Y, f} \right\|_{\mathcal{K}} \cdot \left\| F_{\mp}^{Y, g} \right\|_{\mathcal{K}} < \infty.$$

(2) Setting

$$J := (I \cap [-2a_l, 2a_l]) + [-\delta, \delta]$$

and defining

$$\begin{aligned}
S(\varepsilon) &:= \sup_{\substack{E \in I \cap [-2a_l, 2a_l] \\ X \in [E-\delta, E+\delta]}} |h(E \mp i\varepsilon, X) - h(E, X)| \\
&\leq \sup_{\substack{E \in I \cap [-2a_l, 2a_l] \\ X \in J}} |h(E \mp i\varepsilon, X) - h(E, X)|,
\end{aligned}$$

by Lemma 3.1.6(b), we have

$$\lim_{\varepsilon \downarrow 0} S(\varepsilon) = 0.$$

Using that

$$\begin{aligned}
\int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX &= \int_{-\delta}^{\delta} \frac{\varepsilon}{X^2 + \varepsilon^2} dX \\
&= \int_{-\delta\varepsilon^{-1}}^{\delta\varepsilon^{-1}} \frac{1}{X^2 + 1} dX \leq \pi,
\end{aligned} \tag{5}$$

we then get

$$\begin{aligned}
& |B_\delta(\varepsilon)| \\
&= \left| \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} [h(E \mp i\varepsilon, X) - h(E, X)] dX dE \right| \\
&= \left| \frac{1}{\pi} \int_{I \cap [-2a_l, 2a_l]} \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} [h(E \mp i\varepsilon, X) - h(E, X)] dX dE \right| \\
&\leq S(\varepsilon) \cdot \int_{I \cap [-2a_l, 2a_l]} \frac{1}{\pi} \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX dE \\
&\leq S(\varepsilon) \cdot \lambda_1(I) \xrightarrow{\varepsilon \downarrow 0} 0.
\end{aligned}$$

(3) Setting

$$T(\delta) := \sup_{(E, X) \in M_\delta} |h(E, X) - h(E, E)|,$$

by Lemma 3.1.6(c), we have

$$\lim_{\delta \downarrow 0} T(\delta) = 0.$$

With (5), this yields

$$\begin{aligned}
\sup_{\varepsilon > 0} |C_\delta(\varepsilon)| &= \sup_{\varepsilon > 0} \left| \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} [h(E, X) - h(E, E)] dX dE \right| \\
&\leq T(\delta) \cdot \sup_{\varepsilon > 0} \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX dE \\
&= T(\delta) \cdot \sup_{\varepsilon > 0} \frac{1}{\pi} \int_{I \cap [-2a_l, 2a_l]} \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX dE \\
&\leq T(\delta) \cdot \lambda_1(I) \xrightarrow{\delta \downarrow 0} 0.
\end{aligned}$$

(4) Using (5), we have

$$\begin{aligned}
D_\delta(\varepsilon) &= \frac{1}{\pi} \int_{M_\delta} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} h(E, E) dX dE \\
&= \frac{1}{\pi} \int_{I \cap [-2a_l, 2a_l]} \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} h(E, E) dX dE \\
&= \frac{1}{\pi} \int_{I \cap [-2a_l, 2a_l]} h(E, E) \int_{[E-\delta, E+\delta]} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} dX dE \\
&= \frac{1}{\pi} \int_{I \cap [-2a_l, 2a_l]} h(E, E) \int_{-\delta\varepsilon^{-1}}^{\delta\varepsilon^{-1}} \frac{1}{X^2 + 1} dX dE \\
&= \frac{1}{\pi} \int_{-\delta\varepsilon^{-1}}^{\delta\varepsilon^{-1}} \frac{1}{X^2 + 1} dX \int_{I \cap [-2a_l, 2a_l]} h(E, E) dE \\
&\xrightarrow{\varepsilon \downarrow 0} \int_{I \cap [-2a_l, 2a_l]} h(E, E) dE = \mathcal{I}.
\end{aligned}$$

(b) By summing statement (a) over $l \in \{1, \dots, L\}$ and $\sigma \in \{+, -\}$, we immediately get

$$\begin{aligned} & \frac{1}{\pi} \int_I \int^\mathcal{K} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} \overline{F_\mp^{E \mp i\varepsilon, f}(X)} F_\mp^{E \mp i\varepsilon, g}(X) dX dE \\ & \xrightarrow{\varepsilon \downarrow 0} \int^\mathcal{K} \chi_I(E) \widetilde{\mathcal{F}}_\pm f(E) \widetilde{\mathcal{F}}_\pm g(E) dE. \end{aligned} \quad (6)$$

By Theorem 3.1.5(d), we have

$$\begin{aligned} F_\mp^{E \mp i\varepsilon, g} &= \mathcal{F}_0 \left(g - f_\mp^{E \mp i\varepsilon, g} \right) = \mathcal{F}_0 \left(g - V(H - E \pm i\varepsilon)^{-1} g \right) \\ &= \mathcal{F}_0 \left([(H - E \pm i\varepsilon) - V] (H - E \pm i\varepsilon)^{-1} g \right) \\ &= \mathcal{F}_0 \left((H_0 - E \pm i\varepsilon) (H - E \pm i\varepsilon)^{-1} g \right), \end{aligned}$$

so, by Theorem 3.1.4(c), we get

$$\begin{aligned} F_\mp^{E \mp i\varepsilon, g}(X) &= \mathcal{F}_0 \left((H_0 - E \pm i\varepsilon) (H - E \pm i\varepsilon)^{-1} g \right) (X) \\ &= (X - E \pm i\varepsilon) \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} g \right) (X). \end{aligned}$$

This yields

$$\begin{aligned} & \overline{F_\mp^{E \mp i\varepsilon, f}(X)} F_\mp^{E \mp i\varepsilon, g}(X) \\ &= \overline{(X - E \pm i\varepsilon) \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} f \right) (X)} (X - E \pm i\varepsilon) \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} g \right) (X) \\ &= [(X - E)^2 + \varepsilon^2] \cdot \overline{\mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} f \right) (X)} \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} g \right) (X) \end{aligned}$$

and therefore

$$\begin{aligned} & \frac{1}{\pi} \int_I \int^\mathcal{K} \frac{\varepsilon}{(X-E)^2 + \varepsilon^2} \overline{F_\mp^{E \mp i\varepsilon, f}(X)} F_\mp^{E \mp i\varepsilon, g}(X) dX dE \\ &= \frac{\varepsilon}{\pi} \int_I \int^\mathcal{K} \overline{\mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} f \right) (X)} \mathcal{F}_0 \left((H - E \pm i\varepsilon)^{-1} g \right) (X) dX dE \\ &= \frac{\varepsilon}{\pi} \int_I \langle \mathcal{F}_0(H - E \pm i\varepsilon)^{-1} f | \mathcal{F}_0(H - E \pm i\varepsilon)^{-1} g \rangle dE \\ &= \frac{\varepsilon}{\pi} \int_I \langle (H - E \pm i\varepsilon)^{-1} f | (H - E \pm i\varepsilon)^{-1} g \rangle dE, \end{aligned}$$

using in the last step that, by Theorem 3.1.4(a), the operator $\mathcal{F}_0$ is unitary. Together with (6), this implies the statement. $\square$

The preceding limit calculation will be the key ingredient in proving the following result:

Proposition 3.2.4. *The following hold:*

(a) *One has*

$$\widetilde{\mathcal{F}}_\pm(c_{00}(\mathbb{Z}, \mathbb{C}^L)) \subseteq \mathcal{K}.$$

(b) *The operator H has no singular continuous spectrum.*

(c) For every $f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, one has

$$\langle f | P_{ac}(H) g \rangle = \left\langle \tilde{\mathcal{F}}_{\pm} f \middle| \tilde{\mathcal{F}}_{\pm} g \right\rangle_{\mathcal{K}}.$$

Proof. (a) For $E \in \mathbb{R}$ and $\varepsilon > 0$, one has

$$\begin{aligned} (H - E - i\varepsilon)^{-1} - (H - E + i\varepsilon)^{-1} &= 2i\varepsilon(H - E - i\varepsilon)^{-1}(H - E + i\varepsilon)^{-1}, \\ &= 2i\varepsilon(H - E + i\varepsilon)^{-1}(H - E - i\varepsilon)^{-1}, \end{aligned}$$

so, by Lemma 3.2.3(b), for every compact subset $I \subseteq \mathbb{R} \setminus \mathfrak{D}$ and $f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we get

$$\begin{aligned} & \frac{1}{2\pi i} \int_I \langle f | [(H - E - i\varepsilon)^{-1} - (H - E + i\varepsilon)^{-1}] g \rangle dE \\ &= \frac{1}{2\pi i} \int_I \langle f | 2i\varepsilon(H - E \mp i\varepsilon)^{-1}(H - E \pm i\varepsilon)^{-1} g \rangle dE \\ &= \frac{\varepsilon}{\pi} \int_I \langle (H - E \pm i\varepsilon)^{-1} f | (H - E \pm i\varepsilon)^{-1} g \rangle dE \\ &\xrightarrow{\varepsilon \downarrow 0} \int^{\mathcal{K}} \chi_I(E) \overline{\left(\tilde{\mathcal{F}}_{\pm} f \right)(E)} \left(\tilde{\mathcal{F}}_{\pm} g \right)(E) dE. \end{aligned} \quad (7)$$

On the other hand, for every compact interval $[a, b]$, by Stone's formula, one has

$$\begin{aligned} & \left\langle f \middle| \frac{1}{2} [P_{[a,b]}(H) + P_{(a,b)}(H)] g \right\rangle \\ &= \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_a^b \langle f | [(H - E - i\varepsilon)^{-1} - (H - E + i\varepsilon)^{-1}] g \rangle dE. \end{aligned}$$

If $[a, b] \cap \sigma_{pp}(H) = \emptyset$, we have

$$\frac{1}{2} [P_{[a,b]}(H) + P_{(a,b)}(H)] = P_{[a,b]}(H).$$

By Theorem 2.4(a), we have $\sigma_{pp}(H) \subseteq \mathfrak{D}$, so, for $[a, b] \subseteq \mathbb{R} \setminus \mathfrak{D}$, we get

$$\begin{aligned} \langle f | P_{[a,b]}(H) g \rangle &= \lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_{[a,b]} \langle f | [(H - E - i\varepsilon)^{-1} - (H - E + i\varepsilon)^{-1}] g \rangle dE \\ &= \int^{\mathcal{K}} \chi_{[a,b]}(E) \overline{\left(\tilde{\mathcal{F}}_{\pm} f \right)(E)} \left(\tilde{\mathcal{F}}_{\pm} g \right)(E) dE, \end{aligned} \quad (8)$$

using (7) in the last step.

Now, for every compact subset $I \subseteq \mathbb{R} \setminus \overline{\mathfrak{D}}$, by Lemma A.1, there exists a decreasing sequence of finite unions of compact intervals $(I_n)_{n \in \mathbb{N}}$ in $\mathbb{R} \setminus \overline{\mathfrak{D}}$, such that $I = \bigcap_{n \in \mathbb{N}} I_n$. Then, by Equation (8) and the continuity of the spectral measure of H , for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we get

$$\begin{aligned} \langle f | P_I(H) f \rangle &= \lim_{n \rightarrow \infty} \langle f | P_{I_n}(H) f \rangle = \lim_{n \rightarrow \infty} \int^{\mathcal{K}} \chi_{I_n}(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right)(E) \right|^2 dE \\ &= \int^{\mathcal{K}} \chi_I(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right)(E) \right|^2 dE, \end{aligned} \quad (9)$$

using the Monotone Convergence Theorem in the last step.

Similarly, using Lemma A.2 and Equation (9), one argues that, for every open subset $O \subseteq \mathbb{R} \setminus \overline{\mathfrak{D}}$ and every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, one has

$$\langle f | P_O(H) f \rangle = \int_{\mathcal{K}} \chi_O(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE. \quad (10)$$

Since, by Theorem 2.4(b), the set $\overline{\mathfrak{D}} \supseteq \sigma_{\text{pp}}(H)$ is at most countable, we have

$$P_{\overline{\mathfrak{D}}}(H) = P_{\text{pp}}(H) \quad (11)$$

and therefore

$$P_{\mathbb{R} \setminus \overline{\mathfrak{D}}}(H) = P_{\mathbb{R} \setminus \sigma_{\text{pp}}(H)}(H).$$

For the open set $O = \mathbb{R} \setminus \overline{\mathfrak{D}}$ and $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, using Equation (10), this yields

$$\begin{aligned} \langle f | P_{\mathbb{R} \setminus \sigma_{\text{pp}}(H)}(H) f \rangle &= \langle f | P_{\mathbb{R} \setminus \overline{\mathfrak{D}}}(H) f \rangle \\ &= \int_{\mathcal{K}} \chi_{\mathbb{R} \setminus \overline{\mathfrak{D}}}(E) \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE \\ &= \int_{\mathcal{K}} \left| \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) \right|^2 dE = \left\| \tilde{\mathcal{F}}_{\pm} f \right\|_{\mathcal{K}}^2, \end{aligned} \quad (12)$$

using in the last step that, by Theorem 2.4(b), the set $\overline{\mathfrak{D}}$ is at most countable and therefore a Lebesgue zero-set. This immediately implies $\tilde{\mathcal{F}}_{\pm} f \in \mathcal{K}$.

- (b) Given a Lebesgue zero-set $C \subseteq \mathbb{R} \setminus \overline{\mathfrak{D}}$, by the outer regularity of the Lebesgue measure, there exists a decreasing sequence of open subsets $(C_n)_{n \in \mathbb{N}}$ of $\mathbb{R} \setminus \overline{\mathfrak{D}}$ with $C \subseteq C_n$, such that

$$\lim_{n \rightarrow \infty} \lambda_1(C_n) = \lambda_1(C) = 0.$$

Then, by the Monotone Convergence Theorem and Equation (10), for every element $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we get

$$\begin{aligned} 0 \leq \langle f | P_C(H) f \rangle &= \lim_{n \rightarrow \infty} \langle f | P_{C_n}(H) f \rangle \\ &= \lim_{n \rightarrow \infty} \int_{\mathcal{K}} \chi_{C_n}(E) \overline{\left(\tilde{\mathcal{F}}_{\pm} f \right) (E)} \left(\tilde{\mathcal{F}}_{\pm} f \right) (E) dE = 0 \end{aligned}$$

and therefore $P_C(H) = 0$ since $c_{00}(\mathbb{Z}, \mathbb{C}^L) \subseteq \ell^2(\mathbb{Z}, \mathbb{C}^L)$ is dense. This proves that H has purely absolutely continuous spectrum on $\mathbb{R} \setminus \overline{\mathfrak{D}}$. Together with Equation (11), this implies the absence of singular continuous spectrum for the operator H .

- (c) By (b), we have

$$P_{\mathbb{R} \setminus \sigma_{\text{pp}}(H)}(H) = P_{\text{ac}}(H),$$

which together with Equation (12) yields

$$\langle f | P_{\text{ac}}(H) f \rangle = \langle f | P_{\mathbb{R} \setminus \sigma_{\text{pp}}(H)}(H) f \rangle = \left\| \tilde{\mathcal{F}}_{\pm} f \right\|_{\mathcal{K}}^2 = \left\langle \tilde{\mathcal{F}}_{\pm} f \middle| \tilde{\mathcal{F}}_{\pm} f \right\rangle_{\mathcal{K}}$$

for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$. By polarization, this implies

$$\langle f | P_{\text{ac}}(H) g \rangle = \left\langle \tilde{\mathcal{F}}_{\pm} f \middle| \tilde{\mathcal{F}}_{\pm} g \right\rangle_{\mathcal{K}} \quad \forall f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L). \quad \square$$

With this preparation, by adapting the ideas in the proof of [AW21, Thm. 4.3.1] to our model, we can finally show that, similar to $\mathcal{F}_0$, the operators $\tilde{\mathcal{F}}_\pm$ give rise to unitary operators from $\ell^2(\mathbb{Z}, \mathbb{C}^L)$ to $\mathcal{K}$:

Theorem 3.2.5. *The maps $\tilde{\mathcal{F}}_\pm|^\mathcal{K}$ extend to bounded linear operators*

$$\mathcal{F}_\pm : \ell^2(\mathbb{Z}, \mathbb{C}^L) \rightarrow \mathcal{K}$$

such that the following hold:

- (a) $\mathcal{F}_\pm^* \mathcal{F}_\pm = P_{\text{ac}}(H)$.
- (b) $\mathcal{F}_\pm H = M_{\text{Id}} \mathcal{F}_\pm$.
- (c) $\mathcal{F}_\pm \mathcal{F}_\pm^* = \mathbf{1}_\mathcal{K}$.

Proof. By Proposition 3.2.4, for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we have

$$\left\| \tilde{\mathcal{F}}_\pm f \right\|_\mathcal{K}^2 = \left\langle \tilde{\mathcal{F}}_\pm f \middle| \tilde{\mathcal{F}}_\pm f \right\rangle_\mathcal{K} = \langle f | P_{\text{ac}}(H) f \rangle = \|P_{\text{ac}}(H) f\|^2 \leq \|f\|^2,$$

which, due to the density of $c_{00}(\mathbb{Z}, \mathbb{C}^L)$ in $\ell^2(\mathbb{Z}, \mathbb{C}^L)$, shows that $\tilde{\mathcal{F}}_\pm|_{c_{00}(\mathbb{Z}, \mathbb{C}^L)}^\mathcal{K}$ extend to linear operators

$$\mathcal{F}_\pm : \ell^2(\mathbb{Z}, \mathbb{C}^L) \rightarrow \mathcal{K}$$

with $\|\mathcal{F}_\pm\| \leq 1$. To show that $\mathcal{F}_\pm$ is an extension of $\tilde{\mathcal{F}}_\pm|^\mathcal{K}$, it remains to show that

$$\mathcal{F}_\pm f = \tilde{\mathcal{F}}_\pm f \tag{13}$$

for every $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$. By construction, Equation (13) holds for every $f \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$. Now, let $f \in \ell^1(\mathbb{Z}, \mathbb{C}^L)$ and consider the sequence $(f_N)_{N \in \mathbb{N}} \subseteq c_{00}(\mathbb{Z}, \mathbb{C}^L)$ defined by

$$f_N(n) = \begin{cases} f(n) & \text{if } |n| \leq N \\ 0 & \text{if } |n| > N. \end{cases}$$

Then, for every $E \in [-2a_l, 2a_l] \setminus \mathfrak{D}$, we have

$$\begin{aligned} \left(\tilde{\mathcal{F}}_\pm f_N \right)_{l,\sigma}(E) &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle w_{l,\pm}^{E,\sigma}(n) \middle| f_N(n) \right\rangle_{\mathbb{C}^L} = \frac{1}{\sqrt{2\pi}} \sum_{n=-N}^N \left\langle w_{l,\pm}^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L} \\ &\xrightarrow{N \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle w_{l,\pm}^{E,\sigma}(n) \middle| f(n) \right\rangle_{\mathbb{C}^L} = \left(\tilde{\mathcal{F}}_\pm f \right)_{l,\sigma}(E). \end{aligned} \tag{14}$$

On the other hand, one has $f_N \xrightarrow{N \rightarrow \infty} f$ in $\ell^2(\mathbb{Z}, \mathbb{C}^L)$ and therefore, since $\mathcal{F}_\pm$ is continuous, we also have $\mathcal{F}_\pm f_N \xrightarrow{N \rightarrow \infty} \mathcal{F}_\pm f$ in $\mathcal{K}$. For $l \in \{1, \dots, L\}$ and $\sigma \in \{+, -\}$, this implies the existence of a subsequence $(f_{N_j})_{j \in \mathbb{N}}$ such that

$$(\mathcal{F}_\pm f_{N_j})_{l,\sigma}(E) \xrightarrow{j \rightarrow \infty} (\mathcal{F}_\pm f)_{l,\sigma}(E)$$

for almost every $E \in [-2a_l, 2a_l]$. Together with (14), this implies that, for almost every $E \in [-2a_l, 2a_l]$, one has

$$(\mathcal{F}_\pm f)_{l,\sigma}(E) = \lim_{j \rightarrow \infty} (\mathcal{F}_\pm f_{N_j})_{l,\sigma}(E) = \lim_{j \rightarrow \infty} \left(\tilde{\mathcal{F}}_\pm f_{N_j} \right)_{l,\sigma}(E) = \left(\tilde{\mathcal{F}}_\pm f \right)_{l,\sigma}(E)$$

and therefore

$$\mathcal{F}_\pm f = \tilde{\mathcal{F}}_\pm f.$$

We now show that the maps $\mathcal{F}_\pm$ satisfy the above statements:

(a) By Proposition 3.2.4, for every $f, g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, we have

$$\langle f | \mathcal{F}_\pm^* \mathcal{F}_\pm g \rangle = \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle_{\mathcal{K}} = \left\langle \tilde{\mathcal{F}}_\pm f \middle| \tilde{\mathcal{F}}_\pm g \right\rangle_{\mathcal{K}} = \langle f | P_{\text{ac}}(H) g \rangle,$$

which implies the statement since $c_{00}(\mathbb{Z}, \mathbb{C}^L)$ is dense in $\ell^2(\mathbb{Z}, \mathbb{C}^L)$.

(b) This is proven analogously to Theorem 3.1.4(c).

(c) By (a), for $f, g \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$, we have

$$\begin{aligned} \langle \mathcal{F}_\pm f | (\mathcal{F}_\pm \mathcal{F}_\pm^*) \mathcal{F}_\pm g \rangle &= \langle f | (\mathcal{F}_\pm^* \mathcal{F}_\pm) (\mathcal{F}_\pm^* \mathcal{F}_\pm) g \rangle = \langle f | P_{\text{ac}}(H) P_{\text{ac}}(H) g \rangle \\ &= \langle f | P_{\text{ac}}(H) g \rangle = \langle f | \mathcal{F}_\pm^* \mathcal{F}_\pm g \rangle = \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle = \langle \mathcal{F}_\pm f | \mathbf{1}_{\mathcal{K}} \mathcal{F}_\pm g \rangle. \end{aligned}$$

To prove that $\mathcal{F}_\pm \mathcal{F}_\pm^* = \mathbf{1}_{\mathcal{K}}$, it therefore suffices to show that $\mathcal{F}_\pm(\ell^2(\mathbb{Z}, \mathbb{C}^L))$ is a dense subspace of $\mathcal{K}$, which is equivalent to $\mathcal{F}_\pm(\ell^2(\mathbb{Z}, \mathbb{C}^L))^\perp = \{0\}$.

So, let $f \in \mathcal{F}_\pm(\ell^2(\mathbb{Z}, \mathbb{C}^L))^\perp$. Then, for every measurable subset $I \subseteq \mathbb{R}$ and every $g \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$, one has

$$\begin{aligned} 0 &= \langle \mathcal{F}_\pm P_I(H) g | f \rangle = \langle \chi_I \mathcal{F}_\pm g | f \rangle \\ &= \sum_{\substack{l \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \int_{[-2a_l, 2a_l]} \chi_I(E) \overline{(\mathcal{F}_\pm g)_{l, \sigma}(E)} f_{l, \sigma}(E) dE \\ &= \int_I \sum_{\substack{l \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \chi_{[-2a_l, 2a_l]}(E) \overline{(\mathcal{F}_\pm g)_{l, \sigma}(E)} f_{l, \sigma}(E) dE, \end{aligned}$$

using (b) for the second equality. Since this equality holds for every measurable subset $I \subseteq \mathbb{R}$, the integrand must be zero. Thus, for every $g \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$, one has

$$0 = \sum_{\substack{l \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \chi_{[-2a_l, 2a_l]}(E) \overline{(\mathcal{F}_\pm g)_{l, \sigma}(E)} f_{l, \sigma}(E) = \sum_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} \overline{(\mathcal{F}_\pm g)_{l, \sigma}(E)} f_{l, \sigma}(E)$$

for almost every $E \in \mathbb{R}$ (see Definition 2.2). We now consider specific functions of the form $g = \delta_n e_k$ with $n \in \mathbb{Z}$ and $k \in \{1, \dots, L\}$. Using that

$$\overline{(\mathcal{F}_\pm(\delta_n e_k))_{l, \sigma}(E)} = \overline{w_{l, \pm}^{E, \sigma}(n) e_k} = \left\langle e_k \middle| w_{l, \pm}^{E, \sigma}(n) \right\rangle,$$

for every $n \in \mathbb{Z}$ and $k \in \{1, \dots, L\}$, we get

$$0 = \sum_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} \left\langle e_k \middle| w_{l, \pm}^{E, \sigma}(n) \right\rangle f_{l, \sigma}(E)$$

for almost every $E \in \mathbb{R}$. Since the set

$$\{(n, k) : n \in \mathbb{Z}, k \in \{1, \dots, L\}\}$$

is countable, this implies that, for almost every $E \in \mathbb{R}$, one has

$$0 = \sum_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} \left\langle e_k \middle| w_{l, \pm}^{E, \sigma}(n) \right\rangle f_{l, \sigma}(E) = \left\langle e_k \middle| \sum_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} f_{l, \sigma}(E) w_{l, \pm}^{E, \sigma}(n) \right\rangle$$

for every $n \in \mathbb{Z}$ and $k \in \{1, \dots, L\}$ and therefore

$$0 = \sum_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} f_{l, \sigma}(E) w_{l, \pm}^{E, \sigma}(n) \quad \forall n \in \mathbb{Z},$$

which implies

$$0 = \sum_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} f_{l, \sigma}(E) w_{l, \pm}^{E, \sigma}.$$

By the linear independence of the functions $w_{l, \pm}^{E, \sigma}$ due to Theorem 2.4(d), this implies that, for almost every $E \in \mathbb{R}$, one has

$$f_{l, \sigma}(E) = 0 \quad \forall l \in B^E, \sigma \in \{+, -\}$$

and therefore $f = 0$. □

We finish this section with a result that we will need later to calculate the wave operators:

Proposition 3.2.6. *Let $f \in \mathcal{K}$, $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$ and $E \in [-2a_1, 2a_1] \setminus \mathfrak{D}$. Then, for every null-sequence $(\varepsilon_n)_{n \in \mathbb{N}}$, there exists a subsequence $(\varepsilon_{n_k})_{k \in \mathbb{N}}$ such that*

$$\lim_{k \rightarrow \infty} \int^{\mathcal{K}} \overline{f(E)} (\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_k})^{-1} g)(E) dE = \int^{\mathcal{K}} \overline{f(E)} ((\mathcal{F}_0 - \mathcal{F}_{\pm})g)(E) dE.$$

Proof. By Theorem 3.1.5(d) and (b), the sequence $((H_0 - E \pm i\varepsilon_n)^{-1} g)_{n \in \mathbb{N}}$ is bounded in $\ell^\infty(\mathbb{Z}, \mathbb{C}^L)$ and therefore, the sequence $(V(H_0 - E \pm i\varepsilon_n)^{-1} g)_{n \in \mathbb{N}}$ is bounded in $\ell^1(\mathbb{Z}, \mathbb{C}^L)$ and thus in $\ell^2(\mathbb{Z}, \mathbb{C}^L)$. Then, using that, by Theorem 3.2.5, $\mathcal{F}_{\pm}$ is a bounded operator, also $(\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_n)^{-1} g)_{n \in \mathbb{N}}$ is a bounded sequence in $\mathcal{K}$. By the Banach–Alaoglu Theorem it then has a weakly convergent subsequence $(\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_k})^{-1} g)_{k \in \mathbb{N}}$, i.e. there exists a function $G \in \mathcal{K}$, such that

$$\lim_{k \rightarrow \infty} \int^{\mathcal{K}} \overline{f(E)} (\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_k})^{-1} g)(E) dE = \int^{\mathcal{K}} \overline{f(E)} G(E) dE.$$

This implies the existence of a subsequence $(\overline{f} \mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_{k_m}})^{-1} g)_{m \in \mathbb{N}}$ that converges almost everywhere pointwise to the function $\overline{f} G$ (see [Ru86, Thm. 3.12]), i.e. for every $l \in \{1, \dots, L\}$ and $\sigma \in \{+, -\}$, one has

$$G_{l, \sigma}(E) = \lim_{m \rightarrow \infty} (\mathcal{F}_{\pm} V(H_0 - E \pm i\varepsilon_{n_{k_m}})^{-1} g)_{l, \sigma}(E)$$

for almost every $E \in [-2a_l, 2a_l]$. The statement then follows since, by Theorem 2.4(e), for every $E \in [-2a_l, 2a_l] \setminus \mathfrak{D}$, one has

$$\begin{aligned}
(\mathcal{F}_\pm V(H_0 - E \pm i\varepsilon)^{-1}g)_{l,\sigma}(E) &= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle w_{l,\pm}^{E,\sigma}(n) \middle| V(H_0 - E \pm i\varepsilon)^{-1}g(n) \right\rangle_{\mathbb{C}^L} \\
&= \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle (H_0 - E \mp i\varepsilon)^{-1}Vw_{l,\pm}^{E,\sigma}(n) \middle| g(n) \right\rangle_{\mathbb{C}^L} \\
&\xrightarrow{\varepsilon \downarrow 0} \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} \left\langle v_l^{E,\sigma}(n) - w_{l,\pm}^{E,\sigma}(n) \middle| g(n) \right\rangle_{\mathbb{C}^L} \\
&= ((\mathcal{F}_0 - \mathcal{F}_\pm)g)_{l,\sigma}(E). \quad \square
\end{aligned}$$

4 The wave- and scattering-operators

In this section, we use our previous results to compute the wave and scattering operators. We start with the wave operators:

Theorem 4.1. *One has $\Omega_\pm = \mathcal{F}_\pm^* \mathcal{F}_0$.*

Proof. We have

$$\frac{d}{ds} e^{isH} e^{-isH_0} = e^{isH} (iH - iH_0) e^{-isH_0} = ie^{isH} V e^{-isH_0},$$

so

$$e^{itH} e^{-itH_0} = e^{i0H} e^{-i0H_0} + \int_0^t \frac{d}{ds} e^{isH} e^{-isH_0} ds = \mathbf{1} + i \int_0^t e^{isH} V e^{-isH_0} ds.$$

For $f, g \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$, using that, by Proposition 1.2 and Theorem 3.2.5, we have

$$\Omega_\pm = P_{\text{ac}}(H) \Omega_\pm = \mathcal{F}_\pm^* \mathcal{F}_\pm \Omega_\pm,$$

this yields

$$\begin{aligned}
\langle f | \Omega_\pm g \rangle &= \langle f | \mathcal{F}_\pm^* \mathcal{F}_\pm \Omega_\pm g \rangle = \langle \mathcal{F}_\pm f | \mathcal{F}_\pm \Omega_\pm g \rangle \\
&= \lim_{t \rightarrow \mp\infty} \langle \mathcal{F}_\pm f | \mathcal{F}_\pm e^{itH} e^{-itH_0} g \rangle \\
&= \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle + i \lim_{t \rightarrow \mp\infty} \int_0^t \langle \mathcal{F}_\pm f | \mathcal{F}_\pm e^{isH} V e^{-isH_0} g \rangle ds \\
&= \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle + i \lim_{\varepsilon \downarrow 0} \int_0^{\mp\infty} \langle \mathcal{F}_\pm f | \mathcal{F}_\pm e^{isH} V e^{-isH_0} g \rangle e^{\pm\varepsilon s} ds, \quad (15)
\end{aligned}$$

using [RS79, Sec. XI.6: Lem. 5] in the last step. Further, using that, by Theorem 3.2.5(b), we have

$$\mathcal{F}_\pm e^{isH} = e^{isM_{\text{Id}}} \mathcal{F}_\pm,$$

for the integrand, we get

$$\begin{aligned}
\langle \mathcal{F}_\pm f | \mathcal{F}_\pm e^{isH} V e^{-isH_0} g \rangle e^{\pm \varepsilon s} &= \langle \mathcal{F}_\pm f | e^{isM_{\text{Id}}} \mathcal{F}_\pm V e^{-isH_0} g \rangle e^{\pm \varepsilon s} \\
&= \int_{\mathcal{K}} \overline{(\mathcal{F}_\pm f)(E)} e^{isE} (\mathcal{F}_\pm V e^{-isH_0} g)(E) e^{\pm \varepsilon s} dE \\
&= \int_{\mathcal{K}} \overline{(\mathcal{F}_\pm f)(E)} (\mathcal{F}_\pm V e^{-is(H_0 - E \pm i\varepsilon)} g)(E) dE.
\end{aligned}$$

Interchanging integrals with the Fubini–Tonelli Theorem, this yields

$$\begin{aligned}
&i \int_0^{\mp\infty} \langle \mathcal{F}_\pm f | \mathcal{F}_\pm e^{isH} V e^{-isH_0} g \rangle e^{\pm \varepsilon s} ds \\
&= \int_{\mathcal{K}} \overline{(\mathcal{F}_\pm f)(E)} \left(\mathcal{F}_\pm V \int_0^{\mp\infty} i e^{-is(H_0 - E \pm i\varepsilon)} ds g \right) (E) dE \\
&= \int_{\mathcal{K}} \overline{(\mathcal{F}_\pm f)(E)} (\mathcal{F}_\pm V (H_0 - E \pm i\varepsilon)^{-1} g) (E) dE. \tag{16}
\end{aligned}$$

Now, by Proposition 3.2.6, for $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, there exists a null-sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that

$$\begin{aligned}
&\int_{\mathcal{K}} \overline{(\mathcal{F}_\pm f)(E)} (\mathcal{F}_\pm V (H_0 - E \pm i\varepsilon_n)^{-1} g) (E) dE \\
&\xrightarrow{n \rightarrow \infty} \int_{\mathcal{K}} \overline{(\mathcal{F}_\pm f)(E)} ((\mathcal{F}_0 - \mathcal{F}_\pm)g)(E) dE = \langle \mathcal{F}_\pm f | (\mathcal{F}_0 - \mathcal{F}_\pm)g \rangle.
\end{aligned}$$

Together with Equations (15) and (16), we then get

$$\begin{aligned}
\langle f | \Omega_\pm g \rangle &= \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle + \lim_{\varepsilon \downarrow 0} i \int_0^{\mp\infty} \langle \mathcal{F}_\pm f | \mathcal{F}_\pm e^{isH} V e^{-isH_0} g \rangle e^{\pm \varepsilon s} ds \\
&= \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle + \lim_{n \rightarrow \infty} \int_{\mathcal{K}} \overline{(\mathcal{F}_\pm f)(E)} (\mathcal{F}_\pm V (H_0 - E \pm i\varepsilon_n)^{-1} g) (E) dE \\
&= \langle \mathcal{F}_\pm f | \mathcal{F}_\pm g \rangle + \langle \mathcal{F}_\pm f | (\mathcal{F}_0 - \mathcal{F}_\pm)g \rangle = \langle \mathcal{F}_\pm f | \mathcal{F}_0 g \rangle = \langle f | \mathcal{F}_\pm^* \mathcal{F}_0 g \rangle.
\end{aligned}$$

We therefore have

$$\langle f | \Omega_\pm g \rangle = \langle f | \mathcal{F}_\pm^* \mathcal{F}_0 g \rangle$$

for every $f \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$ and $g \in c_{00}(\mathbb{Z}, \mathbb{C}^L)$, which yields $\Omega_\pm = \mathcal{F}_\pm^* \mathcal{F}_0$ since $c_{00}(\mathbb{Z}, \mathbb{C}^L)$ is dense in $\ell^2(\mathbb{Z}, \mathbb{C}^L)$. $\square$

We now want to compute the scattering operator. To this end, we notice that, by Theorem 2.4(d), we have two bases of the same space $\ell^\infty(H, E)$. It is then natural to ask for the change-of-basis matrix relating them. We shall prove that this matrix is precisely the scattering matrix at energy E (Theorem 4.4). We start with the definition:

Definition 4.2. (cf. [BS26b, TODO, TODO]) Let $E \in \mathbb{R} \setminus \mathfrak{D}$ and let $|B^E|$ denote the cardinality of the set B^E . Then, we define the matrix $S^E \in \mathbb{C}^{2|B^E| \times 2|B^E|}$ by

$$\left(w_{l,+}^{E,\sigma} \right)_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} = \left(w_{l,-}^{E,\sigma} \right)_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} S^E. \tag{17}$$

Next, for $l = 0, \dots, L$, we define

$$I_l := \begin{cases} \mathbb{R} \setminus [-2a_1, 2a_1] & \text{if } l = 0, \\ (-2a_l, -2a_{l+1}) \cup (2a_{l+1}, 2a_l) & \text{if } l = 1, \dots, L-1 \\ (-2a_L, 2a_L) & \text{if } l = L. \end{cases}$$

Then, for $E \in \mathbb{R} \setminus \mathfrak{D}$ and $l = 0, \dots, L$, one has

$$|B^E| = l \quad \Leftrightarrow \quad E \in I_l.$$

In particular, S^E is non-trivial only for $E \in [-2a_1, 2a_1] = \sigma(H_0)$ (Proposition 1.1). We further have the following theorem:

Theorem 4.3. ([BS26b, TODO]) *Let $E \in [-2a_1, 2a_1] \setminus \mathfrak{D}$. Then S^E is unitary.*

Since $\mathfrak{D}$ is a Lebesgue zero-set (Theorem 2.4(b)), this shows that we get a unitary operator

$$\int_{\sigma(H_0)}^{\oplus} S^E dE$$

acting on the Hilbert space

$$\begin{aligned} \int_{\sigma(H_0)}^{\oplus} \mathbb{C}^{2|B^E|} dE &\cong \bigoplus_{l=1}^L L^2(I_l, \mathbb{C}^{2l}) \cong \bigoplus_{l=1}^L \bigoplus_{\substack{j \in \{1, \dots, l\} \\ \sigma \in \{+, -\}}} L^2(I_l, \mathbb{C}) = \bigoplus_{\substack{j \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} \bigoplus_{l=j}^L L^2(I_l, \mathbb{C}) \\ &\cong \bigoplus_{\substack{j \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} L^2\left(\bigcup_{l=j}^L I_l, \mathbb{C}\right) \cong \bigoplus_{\substack{j \in \{1, \dots, L\} \\ \sigma \in \{+, -\}}} L^2([-2a_j, 2a_j], \mathbb{C}) = \mathcal{K}. \end{aligned}$$

The isomorphism between $\mathcal{K}$ and $\int_{\sigma(H_0)}^{\oplus} \mathbb{C}^{2|B^E|} dE$ is explicitly by

$$\Psi : \mathcal{K} \rightarrow \int_{\sigma(H_0)}^{\oplus} \mathbb{C}^{2|B^E|} dE, \quad (\Psi f)(E) = (f_{l,\sigma}(E))_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} \in \mathbb{C}^{2|B^E|}.$$

Hence, we may regard $\int_{\sigma(H_0)}^{\oplus} S^E dE$ as a unitary operator on $\mathcal{K}$. We will now show that this operator is unitarily equivalent to the scattering operator:

Theorem 4.4. *The following hold:*

- (a) $\mathcal{F}_- = \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_+.$
- (b) $S = \mathcal{F}_0^* \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_0.$

Proof. (a) Taking adjoints in Equation (17), we obtain

$$\left(\left\langle w_{l,+}^{E,\sigma} \right| \right)_{\substack{l \in B^E \\ \sigma \in \{+, -\}}} = (S^E)^* \left(\left\langle w_{l,-}^{E,\sigma} \right| \right)_{\substack{l \in B^E \\ \sigma \in \{+, -\}}},$$

which, by Definition 3.2.1 and Theorem 3.2.5, implies

$$\mathcal{F}_+ = \left(\int_{\sigma(H_0)}^{\oplus} (S^E)^* dE \right) \mathcal{F}_- \quad \text{and thus} \quad \mathcal{F}_- = \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_+.$$

(b) By (a) and Theorem 3.2.5(c), we have

$$\mathcal{F}_- \mathcal{F}_+^* = \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_+ \mathcal{F}_+^* = \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right)$$

and therefore, by Theorem 4.1, we get

$$S = \Omega_-^* \Omega_+ = \mathcal{F}_0^* \mathcal{F}_- \mathcal{F}_+^* \mathcal{F}_0 = \mathcal{F}_0^* \left(\int_{\sigma(H_0)}^{\oplus} S^E dE \right) \mathcal{F}_0. \quad \square$$

This theorem, together with Theorem 3.1.4, shows that S^E coincides with the fiber of the scattering operator with respect to the energy E of the free Hamiltonian H_0 , proving that it is, in fact, the scattering matrix.

A Monotone approximations of sets

In this section, we present two lemmas on monotone approximations of sets used in the proof of Proposition 3.2.4:

Lemma A.1. *Let $D \subseteq \mathbb{R}$ be closed and $K \subseteq \mathbb{R} \setminus D$ be compact. Then, there exists a decreasing sequence $(K_n)_{n \in \mathbb{N}}$ such that each $K_n \subseteq \mathbb{R} \setminus D$ is a finite union of closed intervals (including singletons) and such that*

$$K = \bigcap_{n \in \mathbb{N}} K_n.$$

Proof. Since D is closed and $K \subseteq \mathbb{R} \setminus D$ is compact, we have

$$r := \text{dist}(D, K) > 0.$$

We set

$$U_n := \left\{ x \in \mathbb{R} : \text{dist}(x, K) < \frac{r}{2n} \right\}$$

and define C_n to be the set of all connected components of U_n that intersect K . Then every $I \in C_n$ is an open interval, and

$$K \subseteq \bigcup_{I \in C_n} I.$$

Since the intervals in C_n are pairwise disjoint and form an open cover of K , compactness of K implies that C_n is finite. We set

$$\tilde{K}_n := \bigcup_{I \in C_n} \bar{I} \quad \text{and} \quad K_n := \bigcap_{m=1}^n \tilde{K}_m, \quad n \in \mathbb{N}.$$

Then $\tilde{K}_n$ is a finite union of closed intervals. Since finite unions of closed intervals are closed under finite intersections, K_n is a finite union of closed intervals. Further

$$K_n \subseteq \tilde{K}_n \subseteq \bar{U}_n \subseteq \left\{x \in \mathbb{R} : \text{dist}(x, K) \leq \frac{r}{2}\right\} \subseteq \mathbb{R} \setminus D$$

and

$$K \subseteq \bigcap_{n \in \mathbb{N}} K_n = \bigcap_{n \in \mathbb{N}} \tilde{K}_n \subseteq \bigcap_{n \in \mathbb{N}} \bar{U}_n = K. \quad \square$$

Lemma A.2. *Let $O \subseteq \mathbb{R}$ be open. Then, there exists an increasing sequence $(O_n)_{n \in \mathbb{N}}$ such that each $O_n \subseteq O$ is compact and such that*

$$O = \bigcup_{n \in \mathbb{N}} O_n.$$

Proof. The statement holds true for the sets

$$O_n := \left\{x \in O : \text{dist}(x, \mathbb{R} \setminus O) \geq \frac{1}{n}, |x| \leq n\right\}, \quad n \in \mathbb{N}. \quad \square$$

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References

- [APRR24] Austen V., D. Parra, A. Rennie, and S. Richard, *Topological Levinson's theorem in presence of embedded thresholds and discontinuities of the scattering matrix: a quasi-1D example*, Reviews in Mathematical Physics, 2024
- [AW21] Aktosun, T., and R. Weder, “Direct and Inverse Scattering for the Matrix Schrödinger Equation,” Springer, 2021
- [BFGS22] Ballesteros, M., G. Franco, G. Garro, and H. Schulz-Baldes, *Band Edge Limit of the Scattering Matrix for Quasi-One-Dimensional Discrete Schrödinger Operators*, Complex Analysis and Operator Theory **16**, 2022
- [BFNS24] Ballesteros, M., G. Franco, I. Naumkin, and H. Schulz-Baldes, *Levinson theorem for discrete Schrödinger operators on the line with matrix potentials having a first moment*, Communications in Contemporary Mathematics **26**, 2024

- [BS26a] Ballesteros, M. and J. Schöber, *Existence and analytic behaviour of Jost solutions for Schrödinger operators on a one-dimensional lattice with varying spectral multiplicity*
- [BS26b] Ballesteros, M. and J. Schöber, *Limiting absorption principle and scattering matrix for Schrödinger operators on a one-dimensional lattice with varying spectral multiplicity*
- [RS79] Reed, M., and B. Simon, “Methods of Modern Mathematical Physics. Volume III: Scattering Theory,” Academic Press, 1979
- [Ru86] Rudin, W., “Real and Complex Analysis,” 3rd ed., McGraw–Hill, 1986